LOK JAGRUTI UNIVERSITY (LJU)
INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of Computer Engineering (701)

Bachelor of Technology (B.E.) – Semester – III

Course Code:	**017012391**
Course Name:	**Fundamentals of Computer Science using Python - I**
Category of Course:	Engineering Science Course (ESC)
Prerequisite Course:	Computer Programming using C (017012191)

Teaching Scheme				
Lecture (L)	**Tutorial (T)**	**Practical (P)**	**Credit**	**Total Hours**
2	**0**	**6**	**5**	**20**

Syllabus

Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	**Introduction to Python and Jupyter Notebooks**			1 (5%)
	1.1 Basic Data Types -integers, Booleans, floats, strings, type() function, typecasting	---	---	
	1.2 Declaring and using variables, Comments, Reading input from users, printing output	---		
	1.3 Arithmetic Operators: +, -, *, /, **, //, % Logical Operators: and, or, not Membership Operators: in, not in Assignment Operators: =, +=, -=, *=, /=, **=, %= Comparison Operators: ==, !=, >, <, >=, <= Ternary Operators, Operator precedence	Classification of Operators - Arithmetic, Relational, Logical, Assignment, Increment/Decrement, Bitwise, Special, Unary, Binary, and Ternary Operators (017012191-Unit-2.4)		
	1.4 Using Interactive Shell, Editing, saving, and running a script	---		
	1.5 Jupyter Notebooks: Creating, Opening, Saving and Downloading Notebooks, running cells	---		
02	**Conditional Execution and Iterations**			2 (15%)
	2.1 Control Statements: Simple if, if-else, if-elif-else, Nested if	---	---	
	2.2 Loops: For using range function with else, while loop with else			
	2.3 Break, Continue, Pass Statement			
03	**Functions, Scoping and Abstraction**			1 (5%)
	3.1 Declaring, Defining and Invoking a Function, Function Specification	---	---	
	3.2 Function arguments: keyword, default, positional, variable-length			
	3.3 Local v/s Global variables			
04	**Immutable Data Structures**			2 (10%)
	4.1 Strings -immutability, declaring, accessing through for loop, slicing, concatenation, len, capitalize(), enumerate(), isalnum(), isalpha(), islower(), isupper(), lower(), upper(), isnumeric(), find(), index(), split(), strip(), removing punctuation using translate(), count()	---	---	
	4.2 Tuples -immutability, create, assign, access, del, slicing, concatenation, comparing tuples using >, <, ==, count() function			
	4.3 Built-in functions -sorted, reversed, min, max			
05	**Mutable Data Structures**			4 (15%)
	5.1 Lists -mutability, declaring, accessing through for loop, append(), count(), extend(), index(), insert(), pop(), remove(), reverse(), sort()	---		
	5.2 Dictionaries -create, access using for loop, clear(), copy(), fromkeys(), get(), has_key(), items(), keys(), update(), values()			
	5.3 Sets -declaring, accessing, set(), frozenset(), issubset(), issuperset(), union(), intersection(), difference(), symmetric_difference(), copy()			
	5.4 Lambda functions: Map, Reduce, Filter			
06	**Working with Files**			1 (10%)
	6.1 Working with text files, opening files -open in w, r and a modes with open function	---	---	
	6.2 Reading data from files -read(), readline(), readlines(), seek() and tell() functions			
	6.3 Writing data to files -write(), writelines(), closing files			
07	**Modules, Directories**			1 (5%)
	7.1 Modules and Packages -difference between import and from…import	---	---	
	7.2 Python OS Module -getcwd(), chdir(), mkdir(), listdir(), remove(), rmdir()			
08	**Introduction to Object Oriented Programming and Exception Handling**			2 (8%)
	8.1 Abstraction & Encapsulation: Defining classes, attributes, methods, constructors, destructors, objects, generators	---	---	
	8.2 Exception Handling -try, except, finally, custom exceptions			
09	**Advanced OOP Concepts and Introduction to NumPy**			2 (17%)
	9.1 Polymorphism -Overloading and overriding			
	9.2 Inheritance -single, multi-level, Method Resolution Order (MRO), Abstract Class			
	9.3 Arrays -Creating a NumPy ndarray Object, 1D, 2D and 3D Arrays	---	---	
	9.4 Array Indexing, Slicing, Shape, Reshaping, Iteration			
	9.5 Built-in functions -concatenate, array_split, where, sort			
10	**Introduction to Matplotlib**			4 (10%)
	10.1 Basic Methods -Plot() with different markers, colors, and linestyles, show() Describing the plot -xlabel, ylabel, title with different fonts, colors and positions Grids and subplots	---	---	
	10.2 Simple Scatterplots with color, size and alpha Simple Bar charts bar() and barh() Simple Histogram with hist() Simple pie charts with pie along with labelling and coloring			

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Bachelor of Technology (B.E.) – Semester - III

Course Code:	**017013392**
Course Name:	**Full Stack Development with Javascript-1**
Category of Course:	Engineering Science Course (ESC)
Prerequisite Course:	Computer Programming using C(017012191), Introduction to Computer Networks(017013293)

Teaching Scheme				
Lecture (L)	**Tutorial (T)**	**Practical (P)**	**Credit**	**Total Hours**
3	**0**	**6**	**6**	**30**

Syllabus

Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	**Introduction to Web and Web design**			1 (5%)
	1.1 Basics of WWW, HTTP Protocol: Features, Request and Response, URL, Client Server architecture	Web, HTTP and DNS(017013293-Unit-9.2)	---	
	1.2 Concepts of Effective Web Design:Website Planning, Web design issues: Browser Compatibility, Bandwidth and Cache, Display Resolution, Look and Feel of Website, Page Layout and Linking, Designing Effective Navigation	---	---	
02	**Introduction to HTML**			4 (10%)
	2.1 Basics of HTML:HTML Page Structure, Building Blocks of HTML:<html>,<head>,<title>,<body>, Editors(Notepad, Notepad++, Sublime Text, VS Code)	---	---	
	2.2 Data formatting tags:<b>, <i>, <u>, , , <font>, , <hr>, , <sup>, <sub>, <del>, <mark>, <small>, <big>, <ins>, Heading:<h1> to <h6>, Paragraph:<p>, <pre>, Links:<a>, href, target, name attributes, Images:<img>, src, alt attributes, <meta>:name, content	---	---	
	2.3 Lists:<li>, <ul>, <ol>, <dl>, <dt>, <dd>, type attribute, Form:<form>, action & method attributes, <input>: type attrbutes: text, radio, checkbox, submit, button, file,password, reset, date, email <textarea>:rows & cols attributes, <label>, <fieldset>,<legend>,<select>, <option>,<optgroup>, Frame:<frameset>:rows, cols attributes, <frame>: src attribute, Table:<table>, <tr>, <td>, <th>, <thead>, <tbody>, <tfoot>, <caption>	---	DOM, Modeling a document, traversing and modifying a DOM tree, DOM collections (FSD-2-Unit-4.2)	
	2.4 HTML Block and Inline Elements: and	---	DOM, Modeling a document, traversing and modifying a DOM tree, DOM collections (FSD-2-Unit-4.2)	
03	**Advance HTML**			3 (10%)
	3.1 HTML Media: Audio:<audio>,<source>, Video:<video>,<source>, YouTube:<iframe>	---	---	
	3.2 HTML Graphics: SVG: Circle, Rectangle, Rounded Rectangle, Star, Logo, Canvas Basics	---	---	
	3.3 HTML5 Tags: Area, Map, Header, Footer, Aside, Article, Meter	---	---	
04	**Introduction to CSS**			4 (10%)
	4.1 Significance of CSS, Structure of CSS, Syntax and Types of CSS: <style> tag, Inline, Internal, External	---	---	
	4.2 Various CSS Selectors: Id, Class, Pseudo class, Pseudo element, Attribute, Universal, Descendant, Child	---	---	
	4.3 Background Rules: background-color, background-image, background-repeat, background short hand, Color Property, Text Manipulation: text-indent, text-decoration, text-transform, text-align, Fonts: font-family, font-style, font-size, font-variant, font-weight	---	---	
	4.4 Border: border-style, border-width, border-color, border shorthand, Boxes: Margins, Padding, float property	---	---	
05	**Advance CSS**			3 (10%)
	5.1 Style images: Rounded Images, Thumbnail, Responsive Images, Transparent Images, Flip, Gradients: Linear, Radiant, Conic, Transitions: Delay, Duration, Property, Animation: Name, Duration, Delay, Iteration Count, Direction, Tool-tips: display property, Visibility, width, background-color, color, text-align	---	Selectors, Events, JQuery HTML(FSD-2-Unit-6.2)	
	5.2 Buttons: background-color, border, color, padding, text-align, display, font-size, Multiple column: column-count, column-gap, column-rule-style, Pagination: Simple, Active and Hoverable, Border, Pagination Size, Variables Basics	---	Selectors, Events, JQuery HTML(FSD-2-Unit-6.2)	
	5.3 2-D Transformation: translate, rotate, scaleX, scaleY, scale, Media Queries: Syntax, Media Types	---	---	
06	**Introduction to Bootstrap**			2 (10%)
	6.1 What is Bootstrap?, Why use Bootstrap?, Bootstrap setup, Containers: Fixed, Fluid, Typography: Heading, Abbreviation, Lists , Colors: text, background, Grid System, Tables: Striped, Bordered, Hover, Contextual Classes: active, success, warning, danger	---	---	
	6.2 Images: Rounded, Circle, Thumbnail, Alerts: Success, Info, Warning, Danger, Button: Style, Size, Active, Disabled, Spinner, Jumbotron	---	---	
	6.3 Progress Bars: Default, Labels, Stripped, Animated, Multiple, Spinners: Simple, Growing, Button, Pagination: Basic, Active, Disabled, Dropdown: Basic, Divider, Header, Dropup, Collapse: Basic, Accordian	---	---	
07	**Bootstrap5**			3 (10%)
	7.1 Navs: Nav Menu, Tabs, Navbar: Basic, Colored, Brand/Logo, Dropdown, Forms and Buttons, Carousel: How to create, Add Caption, Modal: How to create, Animation, Scrollspy, Flex: Horizontal, Vertical, Justified	---	---	
	7.2 Forms: Stacked form, Textarea, Inline Forms, Plain Text, Color Picker, Select Menus, Check, Radios, Toggle Switch, Form Validation	---	---	
	7.3 Grid system: Classes, Stacked and Horizontal, Small, Medium and Large Grid	Grid System(017013392– Unit-6.1)	---	
08	**Javascript**			_)4 (10%)
	8.1 Basics of Javascript: Client Side Scripting with JS, Overview, Characteristics and Advantages, Internal and External Javascript	---	Node.js file system, URL module(FSD-2-Unit-9.2)	
	8.2 Conditions: If, If…Else, Functions: Syntax, Calling function on some event, Loops: for, while, do…while, Datatype and Variables: Primitive Datatype: String, Boolean Number, Non-primitive Datatype: Object, Array:1-D Array, Inbuilt functions: Math: abs(), ceil(), floor(), sqrt(), pow(), String: charAt(), charCodeAt(), concat(), indexOf(), split(), substr(), substring(), toLowerCase(), toUpperCase(), Date Manipulation	If, Else if, Nested if (017012191 – Unit-3.1), Entry & Exit Controlled Loop(017012191 – Unit-4.1), One Dimensional Array(017012191 – Unit-5.1), Unser Defined Functions(017012191 – Unit-7.2)	Node.js file system, URL module(FSD-2-Unit-9.2)	
	8.3 Pop up Boxes: Alert, Confirm, Prompt	---	Node.js NPM, Callback concepts, Events:Event loop, Event emitter, Upload files(FSD-2-Unit-9.3)	
09	**Advance Javascript**			3 (15%)
	9.1 Document Object Model(DOM): document object, DOM Methods: write(), getElementById() getElementsByClassName(), getElementsByTagName(), Traversing and Modifying DOM Tree: innerHTML, attribute, setting style	Unser Defined Functions(017012191 – Unit-7.2)	Filters, Services, Tables, HTTP (FSD-2-Unit-8.3)	
	9.2 Event Handling with Javascript, Form Processing: How to fetch form elements?, Various Mouse and Key Events: onclick, onblur, onchange, onfocus, onmouseover, onmouseout, onkeyup, onkeydown, onkeypress, onsubmit event of form Event Object and This	<form> & <input>(017013392– Unit-2.2)	Filters, Services, Tables, HTTP (FSD-2-Unit-8.3)	
	9.3 Javascript Validation, Regular Expression: Notations to Create R.E., test() method to compare R.E. & string, R.E. For Specifying Special Character, Length of Field, Numeric Value, Email ID	Unser Defined Functions(017012191 – Unit-7.2)	FSD-2(Unit-3,4,9,10)	
10	**Ecmascript Version 6**			3 (10%)
	10.1 Var,Let and Const Keywords, Template Literals, New Strings Methods: includes(), startsWith(), endsWith()	Unser Defined Functions(017012191 – Unit-7.2)	Filters, Services, Tables, HTTP (FSD-2-Unit-8.3)	
	10.2 Array Destructuring, ES6 Objects: Initializers, Constructor, Constructor Function, Object Destructuring	One Dimensional array(017012191 – Unit-5.1)	Filters, Services, Tables, HTTP (FSD-2-Unit-8.3)	
	10.3 Fat Arrow Functions: for/of loop, Maps, ES6 Default Parameters: Default Parameter Values, Function Rest Parameter	---	---	